

Mollie Conrad, MSc.

Senior Data Scientist with 10+ Years of Experience | ML Engineer | ETL | Python Developer
she/her | Kitchener, ON, Canada | 613-403-1504 | meconrad2@gmail.com
[linkedin.com/in/mollie-conrad-a59248bb/](https://www.linkedin.com/in/mollie-conrad-a59248bb/)

Languages: Python, Kotlin, SQL
Libraries: Pandas, Matplotlib, NumPy, SciPy, StreamLit, SKLearn, TensorFlow, SQLAlchemy, PyMongo, PyTest, OpenProject
Tools: Jupyter, GitLab, BitBucket, Jira, MLFlow, IntelliJ, PyCharm, VSCode, Postman
Databases: MongoDB, Redis, PostgreSQL, MSSQL
DevOps: Docker, Kubernetes, Lens, Rancher, Jenkins
Algorithms: Timeseries Anomaly Detection, Regression, Classification, Decision Trees, Dimension Reduction, Clustering: XGBoost/OPTICS/DBScan, CNNs
Soft Skills: Mentoring, Upskilling, Organization, Collaborating, Technical Documentation, Process and Workflows
Hobbies: Mathematical Knitting Pattern Design (@whiteowlcrochetco), Fitness, Making Soup, Sci-Fi/Fantasy Books

Professional Experience

Stream Systems | Remote, Calgary, AB

Senior Data Scientist | Jan 2023 - March 2026

Stream Systems designs simulation software to optimize processes for mining and industrial sectors, allowing stakeholders to quickly address decision-making challenges. Heavily supported over 10 proof-of-concept customer projects, deployed to production and maintained 2 critical system components and 3 ML/DL models in production, and delivered a custom software solution to a global leading OEM.

- Led the team by establishing documentation standards, writing instructional articles, and onboarding/mentoring new team members at a wide range of experience levels
- Developed, deployed, and tested machine learning models and supporting code in Python and Kotlin using Jupyter, IntelliJ, BitBucket, Jenkins, Rancher, Kubernetes, Docker Desktop, and Postman
- Implemented software engineering best practices such as data validation, unit testing, version control, embedded documentation, structuring with schemas and DTOs, and constructive peer code review
- Comfortable working in a fast-paced, remote environment with significant ambiguity
- Lead Data Scientist and Back-End Python Developer for a custom event analysis solution provided to a global leading Original Equipment Manufacturer (OEM)
- Deployed a unified data ETL and retrieval pipeline to production using Python and Kotlin (included data validation, SQL extraction, transformation, and loading into non-relational databases)
- Developed two TensorFlow CNNs to predict haul truck failures (timing and location) and to predict process cycle times for a haul truck (loading and dumping), allowing an increase in simulation accuracy compared to basic stochastic sampling methods and significantly reducing error rates. Monitored experimentation progress with MLFlow.
- Responsible for the technical aspects of customer data onboarding for 10+ proof-of-concept projects. This included data validation, custom data cleaning solutions, data loading, time-series analysis and clustering, data visualizations, and custom solution development for both internal solution architects (customer service) and customers
- Developed an algorithm to identify seasonal representative dates/shifts by leveraging unsupervised clustering, leading to a significant decrease in simulation error within the product
- Standardized Life-of-Mine planning data for simulation integration
- Rapidly deployed a prototype UI tool to accelerate Solution Architect customer engagements using StreamLit

Praemo | Hybrid, Kitchener, ON

Lead Data Scientist | Aug 2022 – Nov 2022

Praemo's Tech Team developed a cloud-native automated Data Science and Machine Learning engine for high-dimensional industrial time-series data. Owned and delivered 9 proof-of-concept customer projects in under 3 years, helping to secure Praemo's first ARR license and contributing to Series A investment. Delivered measurable customer ROI through early detection of manufacturing recipe drift, machine failures (>3 months advance warning via statistical/ML/DL models), and consumable degradation. Promoted 3 times in under 3 years. Fast-paced, remote environment with significant ambiguity.

- Led a team of high-functioning Senior Data Scientists and multiple co-op students
- Designed and implemented structured hiring and onboarding processes, including interview frameworks and training programs
- Facilitated team operations, including performance reviews, code reviews, and roadmap planning
- Owned end-to-end customer engagements, including solution design, delivery, and stakeholder communication
- Translated complex modelling outputs into clear, actionable insights for non-technical stakeholders
- Implemented software engineering best practices such as data validation, unit testing, version control, documentation, and peer code review
- Owned and maintained several real-time anomaly detection models
- Monitored and debugged production deployments using Kubernetes and internal tooling, driving continuous system improvements
- Applied anomaly detection models to diverse datasets, surfacing critical signals such as telecom network anomalies, fibre sensor irregularities preceding environmental events, and gaps in retail maintenance schedules

Praemo | Remote, Kitchener, ON

Senior Data Scientist | Jul 2021 – Aug 2022

- Led technical onboarding for new customers, owning data ingestion (ETL), validation, and end-to-end solution setup
- Built and deployed real-time, unsupervised anomaly detection models using Python (Pandas, NumPy, SciPy, scikit-learn, TensorFlow)
- Communicated findings and progress through regular customer presentations, translating model outputs into clear, actionable recommendations
- Authored technical documentation and customer-facing reports to support transparency and adoption
- Developed reusable utilities to support emerging use cases, including automated distribution fitting tools
- Improved code reliability and maintainability by implementing automated testing with PyTest
- Monitored and debugged production deployments using Kubernetes and internal tooling, ensuring system performance and stability

Praemo | Remote, Kitchener, ON

Data Scientist | Jan 2020 – Jul 2021

- Built and deployed real-time anomaly detection models, including sensor outage detection and statistical thresholding approaches
- Conducted large-scale exploratory data analysis and R&D on high-dimensional time-series data using Python (Pandas, NumPy, Matplotlib)
- Delivered a high-priority proof-of-concept project for a global OEM, leveraging anomaly detection models to generate actionable insights and contributing to a successful Series A raise
- Mentored and supervised co-op students, guiding development of statistical process control models and an image-based anomaly detection CNN

University of Waterloo

Research Assistant, Teaching Assistant | May 2015 – Sept 2018

- Modelled Spectral data of water and heavy water using Monte Carlo Radiative Transfer Fortran program RATRAN
- Analyzed simulated spectral data using Python, Numpy, Matplotlib
- Ran physics tutorials for 1st year physics and engineering students
- Marked assignments and exams for 1st, 2nd, and 4th year physics and cosmology students
- Answered emails and held office hours for students
- Responsible for editing and implementing marking schemes and problem-set solutions
- Received TA appreciation award in 2017, nominated by professors and students

Education

Master of Science in Astrophysics & Gravitation (MSc.)

University of Waterloo | Sept 2016 - Oct 2018

Computationally simulated the role of water (H_2^{16}O & H_2^{18}O) in the formation of intermediate mass stars. Analysis used spectral data from the NGC 7129 reflection nebula, a Monte Carlo radiative transfer FORTRAN program called "RATRAN", and python.

Thesis: <https://uwspace.uwaterloo.ca/handle/10012/13855>

Article: <https://iopscience.iop.org/article/10.3847/1538-4357/ab612c/pdf>

Media: <https://educationnewscanada.com/article/education/level/university/1/891977/the-origin-of-water-on-planets-in-our-universe.html>

Bachelor of Science with Honours in Physics & Astronomy, Biophysics Minor (BSc.)

University of Waterloo | Sept 2011 - April 2015

Developed a Python algorithm that simulates the assignment of telescope optical fibres to specific galactic targets based on a priority ranking system as well as proposed system restrictions.

Thesis: <https://www.linkedin.com/in/mollie-conrad-msc-a59248bb/overlay/51053930/single-media-viewer/>

Professional Development

Aug 2024	Udemy: Introduction to Kotlin
Jan 2023	Coursera Guided Project: Introduction to Relational Database and SQL
Oct 2022	Coursera: Get Up and Running with Typescript
May 2022	DeepLearning.ai: TensorFlow Developer Professional Certificate
May 2022	DeepLearning.ai: Deep Learning Specialization
May 2022	Coursera Guided Project: TensorFlow for CNNs: Learn and Practice CNNs
Apr 2021	University of Colorado via Coursera: Inclusive Leadership: The Power of Workplace Diversity
Feb 2021	Coursera Guided Project: Anomaly Detection in Time Series Data with Keras
Jan 2021	Coursera Guided Project: Getting Started with Tensorflow 2
Apr 2020	CENG: Docker & Kubernetes Overview - Labs
Oct 2019	IBM via Coursera: Data Science Professional Certificate

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